University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of Computer Science and Engineering

Term Final Examination, Autumn 2025

Course Title: Electronic Device and Circuits

Course Code: EEE 0713124

Marks: 50 **Time: 3 Hours**

Answer all questions

1. a) Briefly describe construction, operating principle and characteristics of Junction Field Effect Transistor (JFET) with necessary diagrams. Write the differences between Bipolar Junction Transistor (BJT) and Junction Field Effect Transistor (JFET). **[04]**

any 1 diag

b) Determine the range of values of V_i that will maintain the Zener diode of figure. 1(b) in the "on" state. **[06]**

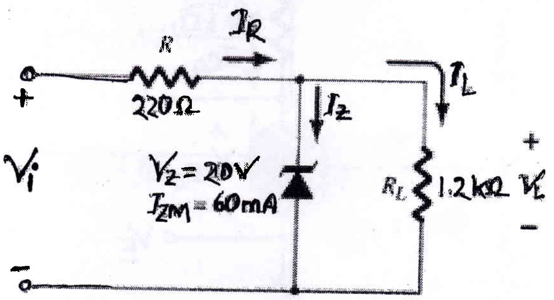

Figure 1: Circuit for 1(b)

2. a) Briefly describe construction, operating principle and characteristics of Depletion type MOSFET. Write the similarities between D-MOSFET and JFET. **[05]**

b) In a transistor amplifier circuit, the current gain plays a vital role in determining its performance. Define the current gain factors α and β of a Bipolar Junction Transistor (BJT), and derive the mathematical relationship between them based on their operational behavior. **[05]**

Defi?

3. a) Determine the voltage V_{CE}, V_C and current I_C for the circuit shown in figure. 3(a). **[05]**

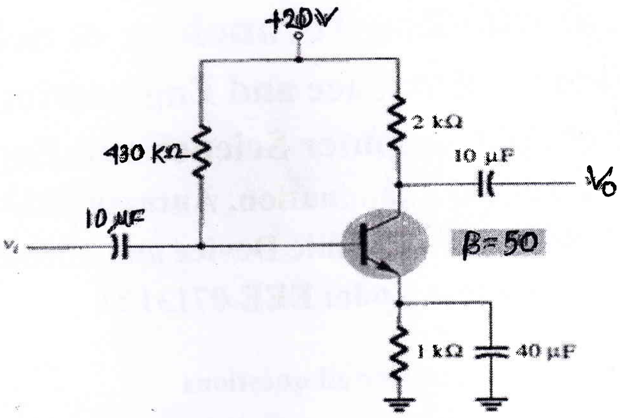

Figure 3: Circuit for 3(a)

b) Determine the voltage V_{CE} and current I_C for the circuit shown in figure. 3(b). **[05]**

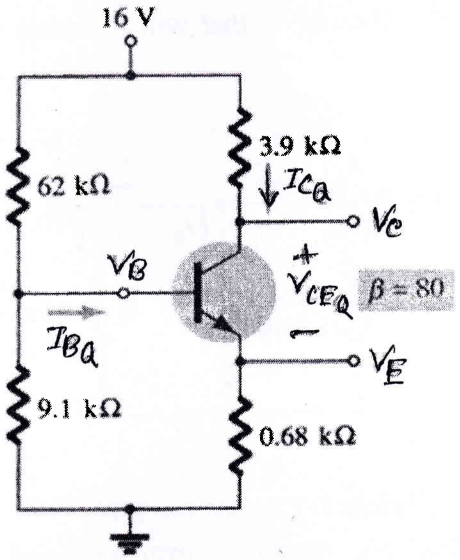

Figure 3: Circuit for 3(b)

4\. a) Sketch transfer curve of a JFET defined by I_{DSS}=20 mA and V_P= -8V. **[05]**

b) Determine the voltage V_{GS}, V_{DS}, V_G, V_S,V_D and current I_D for the circuit shown in figure 4(b). **[05]**

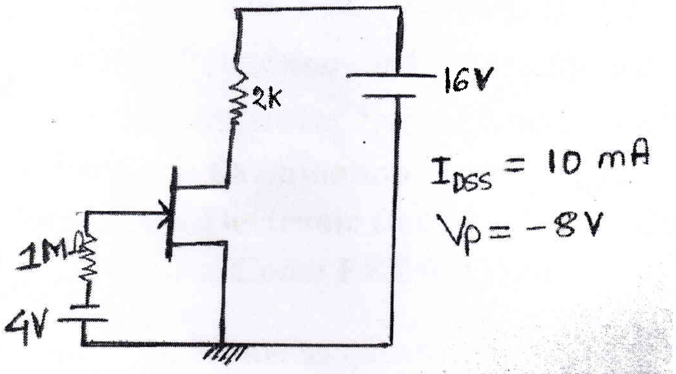

Figure 4: Circuit for 4(b)

5. a) Determine the I_C and V_{CE} for the circuit shown in figure 5(a). Identify the operating mode of the transistor and justify your answer. **[07]**

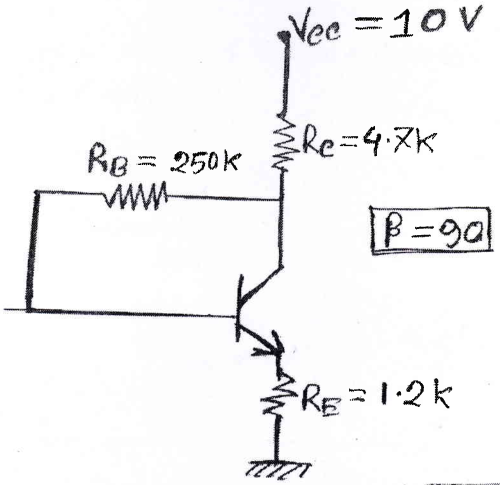

Figure 5: Circuit for 5(a)

b) The NPN transistor is commonly used for signal amplification. **Describe** its construction and explain its operating principle, highlighting how current flows and how the transistor functions in this context. **[03]**